Chapter 5
Complex Differentiation and The Cauchy-Riemann Equation

Analytic Functions:

If a single valued function $f(z)$ is differentiable i.e. $f'(z)$ exists at every point of a domain *D* except possibly at a finite number of exceptional points, then the function is said to be **analytic** in the domain *D*. These exceptional points at which $f'(z)$ does not exist are called **singular points** or **singularities of the function**.

Necessary conditions for $f(z)$ to be analytic:

Rectangular Form:

If $z = x + iy$ and $f(z) = u(x,y) + iv(x,y)$ satisfies the **Cauchy-Riemann equations(C-R)** i.e.,

$$u_x = v_y \text{ and } u_y = -v_x$$

$$\text{i.e., } \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

then $\boldsymbol{f(z)}$ is said to be **analytic.**

Hence**,** at points where $f'(z)$ exists may be obtained from either of

$$f'(z) = u_x + iv_x \text{ or } f'(z) = v_y - iu_y.$$

Polar Form:

If $z = re^{i\theta}$ and $f(z) = u(r,\theta) + iv(r,\theta)$ satisfies the **Cauchy-Riemann equations(C-R)** i.e.

$$u_r = \frac{1}{r}v_\theta \text{ and } v_r = -\frac{1}{r}u_\theta$$

$$\text{i.e., } \frac{\partial u}{\partial r} = \frac{1}{r}\frac{\partial v}{\partial \theta} \text{ and } \frac{\partial v}{\partial r} = -\frac{1}{r}\frac{\partial u}{\partial \theta}$$

then $\boldsymbol{f(z)}$ is said to be **analytic.**

Hence, at points where $f'(z)$ exists may be obtained from either of

$$f'(z) = e^{-i\theta}(u_r + iv_r)$$

$$f'(z) = \frac{e^{-i\theta}}{r}(v_\theta - iu_\theta)$$

Important Formulae:

$\sin at = \dfrac{e^{iat} - e^{-iat}}{2i}$	$\sinh at = \dfrac{e^{at} - e^{-at}}{2}$
$\cos at = \dfrac{e^{iat} + e^{-iat}}{2}$	$\cosh at = \dfrac{e^{at} + e^{-iat}}{2}$
$\cos iay = \cosh ay$	$\sin iay = i\sinh ay$

Example 5.1

Verify C-R equations for the function $\boldsymbol{f(z) = e^{z^2}}$ and hence find $\frac{df}{dz}$ or $f'(z)$.

Solution: Given,

$f(z) = e^{z^2} = e^{(x+iy)^2}$

or $u + iv = e^{x^2-y^2+i\,2xy}$

or $u + iv = e^{x^2-y^2}\, e^{i\,2xy}$

or $u + iv = e^{x^2-y^2}(\cos 2xy + i \sin 2xy)$

or $u + iv = e^{x^2-y^2}\cos 2xy + i\, e^{x^2-y^2}\sin 2xy$

Here, $u(x,y) = e^{x^2-y^2}\cos 2xy$ and $v(x,y) = e^{x^2-y^2}\sin 2xy$.
Now, partially differentiating u and v with respect to x and y, we get
$u_x = 2x\, e^{x^2-y^2}\cos 2xy - 2y\, e^{x^2-y^2}\sin 2xy$
$u_y = -2y\, e^{x^2-y^2}\cos 2xy - 2x\, e^{x^2-y^2}\sin 2xy$
$v_x = 2y\, e^{x^2-y^2}\cos 2xy + 2x\, e^{x^2-y^2}\sin 2xy$
$v_y = -2y\, e^{x^2-y^2}\sin 2xy + 2x\, e^{x^2-y^2}\cos 2xy$

From the above result, we can write

$$u_x = v_y \quad \text{and} \quad v_x = -u_y$$

Since $f(z)$ satisfies Cauchy-Riemann equations, so $f(z)$ is analytic.

$$\begin{aligned}
f'(z) &= u_x + i\, v_x \\
&= 2xe^{x^2-y^2}\cos 2xy - 2y\, e^{x^2-y^2}\sin 2xy + i(2ye^{x^2-y^2}\cos 2xy + 2x\, e^{x^2-y^2}\sin 2xy) \\
&= 2x\, e^{x^2-y^2}(\cos 2xy + i\, \sin 2xy) + i\, 2y\, e^{x^2-y^2}(\cos 2xy - \frac{1}{i}\sin 2xy) \\
&= 2x\, e^{x^2-y^2}(\cos 2xy + i\, \sin 2xy) + i\, 2y\, e^{x^2-y^2}(\cos 2xy + \frac{i^2}{i}\sin 2xy) \\
&= 2x\, e^{x^2-y^2}(\cos 2xy + i\, \sin 2xy) + i\, 2y\, e^{x^2-y^2}(\cos 2xy + i\sin 2xy) \\
&= 2x\, e^{x^2-y^2}e^{i\,2xy} + i\, 2y\, e^{x^2-y^2}e^{i\,2xy} \\
&= 2(x + i\, y)\, e^{x^2-y^2+i\,2xy} = 2(x + i\, y)e^{(x+iy)^2} = 2z\, e^{z^2}.
\end{aligned}$$

Example 5.2

Verify C-R equations for the function $f(z) = z^5$ and hence find $\frac{df}{dz}$ or $f'(z)$.

Solution:

Given

$$f(z) = z^5$$

or, $u + iv = (r\, e^{i\theta})^5$
or, $u + iv = r^5 e^{i\,5\theta}$
or, $u + iv = r^5(\cos 5\theta + i\sin 5\theta)$
or, $u + iv = r^5\cos 5\theta + i\ r^5\sin 5\theta$

Here , $u = r^5\cos 5\theta$ and $v = r^5\sin 5\theta$
Partially differentiating u and v with respect to r and θ, we get
$u_r = 5r^4\cos 5\theta$

$$v_r = 5r^4\sin 5\theta$$

$u_\theta = -5r^5\sin 5\theta$
$v_\theta = 5r^5\cos 5\theta$
From the above result, we can write

$u_r = \frac{1}{r}v_\theta$ and $v_r = -\frac{1}{r}u_\theta$.

Since $f(z)$ satisfies Cauchy-Riemann equations, so $f(z)$ is an analytic function.

$$\begin{aligned}
f'(z) &= e^{-i\theta}(u_r + i\, v_r) \\
&= e^{-i\theta}(5r^4\cos 5\theta + i5r^4\sin 5\theta) \\
&= 5r^4e^{-i\theta}(\cos 5\theta + i\sin 5\theta) \\
&= 5r^4e^{-i\theta}\ e^{i\,5\theta} \\
&= 5r^4e^{i\,4\theta} \\
&= 5(r\, e^{i\,\theta})^4 \\
&= 5\, z^4.
\end{aligned}$$

- **Physical example of complex function and application of Cauchy-Riemann Equation**

- Fluid Flow (Potential Flow in 2D Incompressible, Irrotational Fluids)

$$F(z) = \phi(x, y) + i\psi(x, y)$$

Where

$\phi(x, y) =$ Velocity potential

$\psi(x, y) =$ Stream function

The Cauchy–Riemann equations ensure:

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \qquad \frac{\partial \phi}{\partial y} = -\frac{\partial \psi}{\partial x}$$

which basically means the flow is **incompressible** and **irrotational**.

Physical example: Flow past a cylinder

Take the complex potential

$$F(z) = U(z + \frac{a^2}{z})$$

where $U$is the free stream velocity, a is the cylinder radius, and $z = x + iy$.

- Real part: ϕ gives the **potential field** around the cylinder.
- imaginary part: ψ gives the **streamlines**, which beautifully match real fluid flow patterns in wind tunnels.

This is used in **aerodynamics**, **marine engineering**, and **hydraulic structure design**, e.g., to model flow around bridge piers or ship hulls.

- Electrostatics / Magnetostatics
  In 2D electrostatics, if you consider a region with no free charges, the potential function $\phi(x, y)$satisfies **Laplace's equation**:

$$\nabla^2 \phi = 0$$

Any **analytic complex function** $F(z)$ gives you a solution, because its real and imaginary parts are **harmonic conjugates** (they satisfy Laplace's equation and Cauchy–Riemann).

Physical example: Line charge or conducting cylinder

$$F(z) = -\frac{\lambda}{2\pi\epsilon_0}\ln z$$

- Real part: $\phi = -\frac{\lambda}{2\pi\epsilon_0}\ln r$ is the electric potential.
- Imag part: $\psi = -\frac{\lambda}{2\pi\epsilon_0}\theta$ corresponds to equipotentials or flux lines.

Similar trick works for magnetic fields in 2D, using complex functions to represent vector fields around conductors.

Used in **design of capacitors**, **electric field shielding**, **waveguide cross-sections**, and **transformer cores**.

- Heat Conduction & Steady State Temperature
  If there's no heat source in a 2D region, the **temperature** $T(x, y)$also satisfies Laplace's equation.
  So, if you pick an analytic function $F(z)$:
  $$F(z) = T(x, y) + i\psi(x, y)$$
  then T can model the temperature distribution, and ψ is a conjugate harmonic function with no direct physical meaning but helps solve boundary value problems neatly.

Physical example: Cooling fin problem

A metal plate with complex geometry can have its steady-state temperature mapped by conformally transforming the domain to something simple (e.g., upper half-plane), using $F(z)$, solving the simpler case, and mapping back.

This is used in **thermal engineering**, e.g., heat sinks, microchip cooling, or metallurgical processes.

- Aerodynamics – Airfoils (Joukowsky Transformation)

  The Joukowsky transform:

$$w = z + \frac{1}{z}$$

maps circles in the z-plane to airfoil-like shapes in the w-plane.

By picking a simple flow potential around a circle (which we know analytically), and applying this conformal map, engineers can model **flow around complex airfoil shapes**.

This was historically the key to early airplane wing design, before CFD existed. The Cauchy–Riemann equations guarantee that the transformation preserves harmonic functions, so the mapped flow still satisfies the physical equations.

- Groundwater Flow & Porous Media

The flow of water through soil (Darcy's law) in 2D steady state is governed by the **same Laplace equation** as electrostatics or fluid flow. Complex potentials are used to:

- Locate wells and estimate flow between them
- Model seepage under dams
- Predict contamination plume movements

Physical example:

$$F(z) = \frac{Q}{2\pi k} \ln z$$

where

Q= discharge,

k= hydraulic conductivity.

Why Cauchy–Riemann conditions matters here

The CR equations

$\phi_x = \psi_y,\ \phi_y = -\psi_x$ (in cartesian coordinates)
or, $\phi_r = \frac{\psi_\theta}{r},\ \psi_r = -\frac{\phi_\theta}{r}$ (in polar coordinates)

ensure:

- The vector fields are irrotational and divergence-free — a physical necessity in steady potential problems.
- The real and imaginary parts are harmonic, making them perfect solutions for Laplace's equation.
- We can use powerful tools like contour integration, conformal mapping, and residue calculus to solve PDEs elegantly.

Example 5.3

Flow near a $\mathbf{90}°$ corner, or a stagnation point is given by the complex function, $\boldsymbol{f(z) = z^2}$.

i. Write the function in the form $f(z) = \phi + i\,\psi$
ii. Find the partial derivatives with respect to all independent variables.
iii. Show whether they satisfy the Cauchy-Riemann conditions or not.
iv. If satisfied, find $f'(z)$ using the partial derivatives and simplify as a function of z.
v. Find the derivative (if possible), $f'(z) = u + i\,v$, where u and v are the velocity components in the parallel direction of the axes of coordinates.
vi. Also show that the potential function, ϕ satisfy the Laplace equation.

Solution:

i. In cartesian variables,

$$\begin{aligned} f(x,y) &= (x+iy)^2 \\ &= x^2 + 2ixy - y^2 \\ &= x^2 - y^2 + 2ixy \\ &= \phi(x,y) + i\psi(x,y) \end{aligned}$$

$\phi(x,y) = x^2 - y^2$
$\psi(x,y) = 2xy$

ii. Partial Derivatives with respect to x and y

$\phi_x = 2x$; $\phi_y = -2y$
$\psi_x = 2y$; $\psi_y = 2x$

iii. As
$\phi_x = \psi_y = 2x$
and
$\psi_x = -\phi_y = 2y$

So, they satisfy the Cauchy-Riemann conditions

iv.

$$f'(z) = \phi_x + i\,\psi_x = 2x + i\,2y = 2(x+iy) = 2z$$

v. From the solution in ii,
Horizontal velocity component (in the direction of $\hat{x}$) is, $u(x,y) = \phi_x = 2x$
And
Vertical velocity component (in the direction of $\hat{y}$) is, $v(x,y) = \phi_y = 2y$

vi. Laplace equation for $\phi(x,y)$ in polar coordinate is

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} = 0$$

Now,
$\phi_{xx} = 2$
$\phi_{yy} = -2$

So
$\nabla^2\phi = 2 - 2 = 0$

So, the potential function, $\phi(x,y)$ satisfy the Laplace equation.

Example 5.4

Joukowsky Transformation for airfoil is given by the complex function, $\boldsymbol{f(z) = z + \frac{1}{z}}$.

i. Write the function in the form $f(z) = \phi + i\,\psi$
ii. Find the partial derivatives with respect to all independent variables.
iii. Show whether they satisfy the Cauchy-Riemann conditions or not.
iv. If satisfied, find $f'(z)$ using the partial derivatives and simplify as a function of z.
v. Find the derivative (if possible), $f'(z) = u + i\,v$, where u and v are the velocity components in the parallel direction of the axes of the flow around the airfoil.
vi. Also show that the potential function, ϕ satisfy the Laplace equation.

Solution:

i. In polar variables,

$$\begin{aligned} f(r,\theta) &= r\,e^{i\theta} + \frac{1}{r\,e^{i\theta}} \\ &= r\cos\theta + i\,r\sin\theta + r^{-1}\cos\theta - i\,r^{-1}\sin\theta \\ &= r\cos\theta + r^{-1}\cos\theta + i(\,r\sin\theta - r^{-1}\sin\theta) \\ &= \phi(r,\theta) + i\psi(r,\theta) \end{aligned}$$

$$\phi(r,\theta) = r\cos\theta + \frac{\cos\theta}{r}$$

$$\psi(r,\theta) = r\sin\theta - \frac{\sin\theta}{r}$$

ii. Partial Derivatives with respect to r and θ

$$\phi_r = \cos\theta - \frac{\cos\theta}{r^2} \quad ; \quad \frac{\phi_\theta}{r} = \frac{1}{r}\left(-\mathrm{r}\sin\theta - \frac{\sin\theta}{r}\right) = -\sin\theta - \frac{\sin\theta}{r^2}$$

$$\psi_r = \sin\theta + \frac{\sin\theta}{r^2} \quad ; \quad \frac{\psi_\theta}{r} = \frac{1}{r}\left(\mathrm{r}\cos\theta - \frac{\cos\theta}{r}\right) = \cos\theta - \frac{\cos\theta}{r^2}$$

iii. As

$$\phi_r = \frac{\psi_\theta}{r} = \cos\theta - \frac{\cos\theta}{r^2}$$

and

$$\psi_r = -\frac{\phi_\theta}{r} = \sin\theta + \frac{\sin\theta}{r^2}$$

So, they satisfy the Cauchy-Riemann conditions

iv.

$$\begin{aligned} f'(z) &= e^{-i\theta}(\phi_r + i\,\psi_r) = \frac{e^{-i\theta}}{r}(\,\psi_\theta - i\,\phi_\theta) \\ &= e^{-i\theta}\left\{\cos\theta - \frac{\cos\theta}{r^2} + i\left(\sin\theta + \frac{\sin\theta}{r^2}\right)\right\} \\ &= e^{-i\theta}(\cos\theta + i\sin\theta) - e^{-i\theta}\left(\frac{\cos\theta}{r^2} - i\frac{\sin\theta}{r^2}\right) \\ &= e^{-i\theta}e^{i\theta} - e^{-i\theta}\left(\frac{\cos\theta}{r^2} - i\frac{\sin\theta}{r^2}\right) \end{aligned}$$

$$= e^{-i\theta}e^{i\theta} - e^{-i\theta}\left(\frac{\cos\theta - i\sin\theta}{r^2}\right)$$
$$= 1 - \frac{e^{-i\theta}e^{-i\theta}}{r^2}$$
$$= 1 - \frac{e^{-2i\theta}}{r^2}$$
$$= 1 - \frac{1}{r^2 e^{2i\theta}}$$
$$= 1 - \frac{1}{(re^{i\theta})^2}$$
$$= 1 - \frac{1}{z^2}$$

v. From the solution in ii,

Radial velocity (in the direction of $\hat{r}$ is, $u(r,\theta) = \phi_r = \cos\theta - \frac{\cos\theta}{r^2}$

And

Tangential velocity (in the direction of $\hat{\theta}$ is, $v(r,\theta) = \frac{\phi_\theta}{r} = \sin\theta - \frac{\sin\theta}{r^2}$

vi. Laplace equation for $\phi(r,\theta)$ in polar co-ordinate is

$$\nabla^2\phi = \frac{\partial^2\phi}{\partial r^2} + \frac{1}{r}\frac{\partial\phi}{\partial r} + \frac{1}{r^2}\frac{\partial^2 T}{\partial\theta^2} = 0$$

Now,

$$\phi_{rr} = \frac{2\cos\theta}{r^3}$$
$$\frac{\phi_{\theta\theta}}{r^2} = \frac{1}{r^2}(-r\cos\theta - \frac{\cos\theta}{r})$$

So

$$\nabla^2\phi = \frac{2\cos\theta}{r^3} + \frac{1}{r}\left(\cos\theta - \frac{\cos\theta}{r^2}\right) + \frac{1}{r^2}\left(-r\cos\theta - \frac{\cos\theta}{r}\right)$$
$$= \frac{2\cos\theta}{r^3} + \frac{\cos\theta}{r} - \frac{\cos\theta}{r^3} - \frac{\cos\theta}{r} - \frac{\cos\theta}{r^3}$$
$$= 0$$

So, the potential function, $\phi(r,\theta)$ satisfy the Laplace equation.

Exercise set: 5.1

1. Write Cauchy-Riemann (C-R) equations in rectangular and polar forms.

2. For the following functions:

$$f(z) = e^{2z}, 2z^2, z^5, \frac{1}{z^3}, z^9$$

(i) separate real and imaginary parts,
(ii) verify **C-R** equations,
(iii) find $\frac{df}{dz}$ or $f'(z)$.

3. Justify whether the following functions satisfy the **(C-R)** equations. If yes, then find $f'(z) = u_x + iv_x$

$$f(z) = \bar{z},\ 2z^2 + 3e^z,\ 2ze^z \text{and } 3z^3 .$$

4. Justify whether the following functions satisfy the **(C-R)** equations. If analytic, then find $f'(z) = e^{-i\theta}(u_r + iv_r)$

$$f(z) = z^2, \frac{1}{z^9}, z^{-\frac{2}{3}} \text{ and } z^{\frac{3}{5}}$$

5. Problems from Book (Complex Analysis with Applications – Dennis G. Zill)

Verify Cauchy-Riemann Equations. If possible, find $f'(z)$: Using Cauchy-Riemann Equation in Cartesian form

a. $f(z) = z^3$
b. $f(z) = 3z^2 + 5z - 6i$
c. $f(z) = z^3 - z$
d. $f(z) = z(z^2 + z)$
e. $f(z) = \bar{z}$
f. $f(z) = i\,\bar{z}$
g. $f(z) = 4z - 6\bar{z} + 3$
h. $f(z) = \bar{z}^2$
i. $f(z) = |z|^2 + i\,z\bar{z}$
j. $f(z) = z + \sin(z)$
k. $f(z) = e^z$
l. $f(z) = 4z^2 + 5z + 9 - i$
m. $f(z) = z^3 + 6z$
n. $f(z) = e^{\bar{z}} = \exp(\bar{z})$
o. $f(z) = e^{-i\,z}$
p. $f(z) = e^{z^2}$
q. $f(z) = e^{-z}$
r. $f(z) = e^{i\,z}$

Using Cauchy-Riemann Equation in Polar form

a. $f(z) = \dfrac{z}{|z|^2}$
b. $f(z) = z^5$
c. $f(z) = \dfrac{1}{z^7}$
d. $f(z) = \dfrac{\bar{z}}{|z|^2}$
e. $f(z) = \ln(z)$
f. $f(z) = z^{1/2}$

6. Physical Application problem

For the following complex functions found in the real-world applications,

i. Write the function in the form $f(z) = \phi + i\,\psi$
ii. Find the partial derivatives with respect to all independent variables.
iii. Show whether they satisfy the Cauchy-Riemann conditions or not.
iv. If satisfied, find $f'(z)$ using the partial derivatives and simplify as a function of z.
v. Find the derivative $f'(z) = u + i\,v$, where u and v are the corresponding horizontal and vertical velocity component of the flow around the airfoil.
vi. Also show that the potential function, ϕ, satisfy the Laplace equation.
    - Flow around a semi-infinite plate: $f(z) = z^{1/2}$
    - Flow around a right corner: $f(z) = z^{2/3}$
    - Uniform flow: $f(z) = z$
    - Flow through a corner, or near a stagnation point: $f(z) = z^2$
    - Flow into a 60° corner: $f(z) = z^3$
    - Flow due to a source doublet: $f(z) = \frac{1}{z}$
    - Flow due to a quadrupole: $f(z) = \frac{1}{z^2}$
    - Line source: $f(z) = \frac{\ln(z)}{2\pi}$
    - Line sink: $f(z) = -\frac{\ln(z)}{2\pi}$
    - Line vortex: $f(z) = \frac{\ln(z)}{2\pi\,i}$
    - Joukowsky transformation for airfoil: $f(z) = z + \frac{1}{z}$
    - Flow past a cylinder of radius R: $f(z) = U\left(z + \frac{R^2}{z}\right)$